

SQL Fundamentals: SELECT and Filtering

1. SELECT *

FROM employees;

id	first-name	last-name	department	salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first-name, last-name

FROM employees

ORDER BY salary DESC;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. **SELECT** id, first-name, last-name, salary
FROM employees
ORDER BY salary **DESC**
LIMIT 3;

id	first-name	last-name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. **SELECT** id, first-name, last-name, department
FROM employees
WHERE department = 'IT';

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. **SELECT** id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance' **AND** salary > 6000

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7. **SELECT** id, first-name, last-name, department
FROM employees
WHERE department = 'HR' **OR** department = 'Marketing'

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id, first-name, last-name, department
FROM employees
WHERE department != 'IT';

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT' 'HR' 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id, first-name, last-name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg